



NM3243 User Experience Design

Final Design Proposal

Design of Scheduling and Video Conferencing System for Students

Group Members	Matriculation Number
Chin Chen Shao, Javier	A0272898E
Joshua Gerard Fabroa Falsado	A0272381E
Pun Ling	A0245189U
Wu Jingya	A0257415A

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Milestone 1: Target User Group

Our target user group is university students engaged in project-based group work. This demographic regularly participates in group projects across academic disciplines, making them highly relevant for a solution that improves collaborative efficiency.

1.1 Problem Identification

Existing research has identified consistent challenges that hinder student collaboration, especially in online environments. According to Roberts and McInnerney (2007) and Ekblaw (2016) (as cited in Bakir et al., 2020), students commonly face issues such as:

- Apathy towards group work and limited understanding of its value
- Lack of collaboration, leadership, or group management skills
- Uneven work distribution and free-riding
- Poor delegation of roles and unclear assessment structures
- Inequality in group members' skill levels

These challenges are further intensified in online settings, where communication is more fragmented and coordination becomes more difficult (Kemp & Grieve, 2014, as cited in Bakir et al., 2020). Compounding these issues is the reality of students' busy and conflicting schedules, which often delay meetings and overall reduce meaningful engagement.

1.1.1 Scheduling as a Key Barrier

One of the biggest challenges students face is finding time to meet. Coordination typically happens through group chats, resulting in excessive back-and-forth messaging, frequent rescheduling, and growing frustration as group size increases. This problem stems from two main issues: the absence of a centralized scheduling system and the lack of integration with university timetables. Most tools require manual input of availability, which is both tedious and prone to error.

1.1.2 Unproductivity and Chaotic Discussions

Even when meetings are successfully arranged, they often lack structure and flow. Ruhleder and Jordan (2001) observed that online meetings frequently suffer from speaker overlap, awkward pauses, and misinterpretation of conversational cues, such as mistaking a pause for the end of a thought or missing subtle non-verbal signals. Participants also struggled to gain each other's attention, sometimes needing to interrupt ongoing side conversations. These communication breakdowns result in chaotic, inefficient meetings that hinder overall group progress.

1.2 Current Solutions and Gaps to Address

Existing tools fall short for the needs of university students:

- Calendar apps (Google, Apple, Outlook) do not sync with school timetables or support group-wide availability checks.
- Doodle/When2Meet require manual input and are not ideal for recurring meetings; mobile usability is limited.
- University tools (e.g., NUSMods) support individual planning but not group coordination or external commitments.

Furthermore, existing tools lack a seamless transition between scheduling and conducting meetings. To address this gap, our goal is to develop an integrated meeting host system that streamlines the scheduling of group meetings and facilitates more productive meetings for university students. This system aims to mitigate the problems of difficulties in coordinating meeting time with group mates and unproductive meetings caused by awkward silences or chaotic discussions.

Milestone 2: Data Gathering

To better understand our target user group, our goal was to uncover pain points, existing strategies and unmet needs of university students guided by the following high-level questions:

1. How prevalent is the problem among our target user group?
2. What systems, methods or processes do users currently use to schedule and carry out group meetings?
3. What pain points or difficulties do they face when scheduling or hosting meetings?
4. Which features or attributes of the current systems do they value?
5. What improvements do they wish to see in new systems that will help them better schedule and host meetings?

To do so, we adopted a mixed-methods approach comprising of both semi-structured interviews and online surveys to gather in-depth qualitative insights and broad quantitative trends.

2.1 Semi-Structured Interviews

We selected semi-structured interviews as our primary qualitative method due to its balance of structure and flexibility. This approach ensured a common structure across interviews while allowing interviewers to explore individual experiences in greater depth which was particularly useful given that there were multiple interviewers involved. Each session began with obtaining informed consent, followed by a brief explanation of the purpose of the interview. Audio recordings of each interview were collected exclusively for the purposes of data analysis and accurate transcription, in accordance with ethical research practices. The interview questions, developed based on our high-level research questions, are provided in Appendix A.

2.2 Online Questionnaires

We utilized online questionnaires via Google Forms to reach a broader student population efficiently. This approach enabled us to quantify the prevalence and severity of common pain points, capture a wide range of current practices and preferences and collect both structured (quantitative) and open-ended (qualitative) responses. Moreover, surveys are valuable for identifying trends and validating insights from our interviews. The complete list of survey questions can be found in Appendix B.

Milestone 3: Data Analysis

For data analysis, our group constructed an overarching high-level question as well as a hypothesis to determine the results we aimed to gather.

Question: “What pain points or difficulties do students face when scheduling or hosting meetings?”

Hypothesis: “Student’s find the process of organizing and facilitating meetings too disorganized and unproductive at times due to their hectic and varying schedule/lifestyles”

Through these anchor points, we sought to identify any glaring needs that might have been unmet with current systems in the whole experience of scheduling and hosting meetings, as well as requesting brief feedback of our proposed web application to guide our eventual design process.

3.1 Questionnaire - Context and Results

Our questionnaire was found useful in confirming our group’s hypothesis that university students find scheduling to be tedious, providing highly useful quantitative data as well as some qualitative insights.

3.1.1 Questionnaire - Context

For our questionnaire, a total of 41 responses were recorded, with 80% of them in their second year of university and above, suggesting that they have plenty of experience with group meetings. This is further confirmed where 51.2% of respondents indicated they have meetings a few times a month, while 31.7% meet as often as a few times every week. Interestingly, no one indicated that they never have group meetings.

A vast majority of students indicated ‘Class Projects’ as one of their uses for these group meetings at a staggering 97.6%. Thus, we concluded that the respondents for the questionnaire largely reflect our target demographic of Students in Universities frequently attending meetings.

3.1.2 Questionnaire - Results

On average, how many participants are there in each group meeting you've attended?

 [Copy chart](#)

41 responses

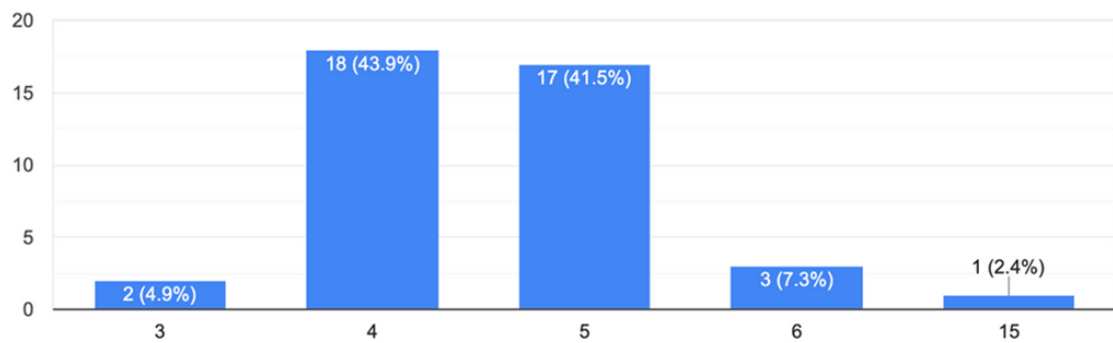


Fig 3.1: Number of Participants in Group Meetings

How long does it usually take to decide on a time to meet?

41 responses

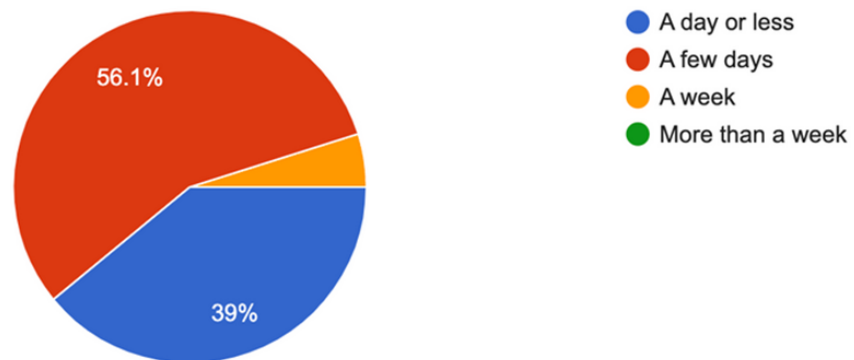


Fig 3.2: How long it takes for a group meeting to be decided

Various valuable insights guided our design process through the use of this questionnaire. Some highlights are firstly, when asked on average how many participants there are in each group meeting, a combined 85.4% indicated that their meetings typically involved 4-5 people (Fig 3.1). Furthermore, 56.1% of respondents average their time to decide on a meeting date to be a few days.

Current systems and pain points were further explored, where we found that 97% of respondents currently utilize the online group messaging platform Telegram to discuss meeting times. We found that 95.1% of respondents find it tedious to find common free time in everyone's schedule, and 97.6% indicated that meetings are usually for planning/discussions rather than real-time collaboration on a specific task.

The most important finding was that 70.7% of respondents found Awkward Silences to be a significant problem that they encounter when participating and attending group meetings. This key statistic greatly influenced our eventual design choice of implementing a unique feature that aims to combat such awkward silences.

From this data, it was concluded that university students do indeed find the experience of scheduling a meeting to be tedious.

3.2 Semi-structured Interviews - Context and Results

The issue of disorganized group meetings was shown to be more prevalent as achieved by more nuanced insights found by our semi-structured interviews, which placed great emphasis on finding out the user's experience in hosting and participating in the meetings themselves rather than scheduling.

3.2.1 Semi-structured Interviews - Context

A total of 5 interviews were conducted, where our group reached out to university students from different academic years and faculties of study. Each interviewee received and signed a consent form to allow for the recording and sharing of their responses (Appendix A).

3.2.2 Semi-structured interviews - Results

Some key insights from our interviews were firstly, our questionnaire finding of meetings taking a few days to decide on a meeting date, where a common response from interviewees was that it typically takes 2-3 days for team members to indicate availability. It was also found that Zoom was the preferred video conferencing platform due to familiarity and lack of technical issues. Interviewees highlighted however that controlling and giving rights for screen-share can be troublesome.

The most surprising finding was the informal nature of scribing. Our interviewees commonly found that scribing is rather informal and had a lack of structure. For example, no one would be appointed a dedicated scribe and situations arise where either everyone scribed, or no one did at all. As a result, it was found that interviewees would miss out on certain points that could have been helpful if raised during the meeting itself.

Thus, our semi-structured interviews were useful to provide deeper and more qualitative insights that guided our design process along. In all, through our findings the most glaring needs that shaped our design are as follows;

- 1. Easier and Hassle-Free Scheduling**
- 2. More Productive and Structured Meetings**
 - a. Keeping Track of Tasks**
 - b. Scribing Meetings**
 - c. Engaging Discussions to prevent Awkward Silences**

Milestone 4: Requirements Definition

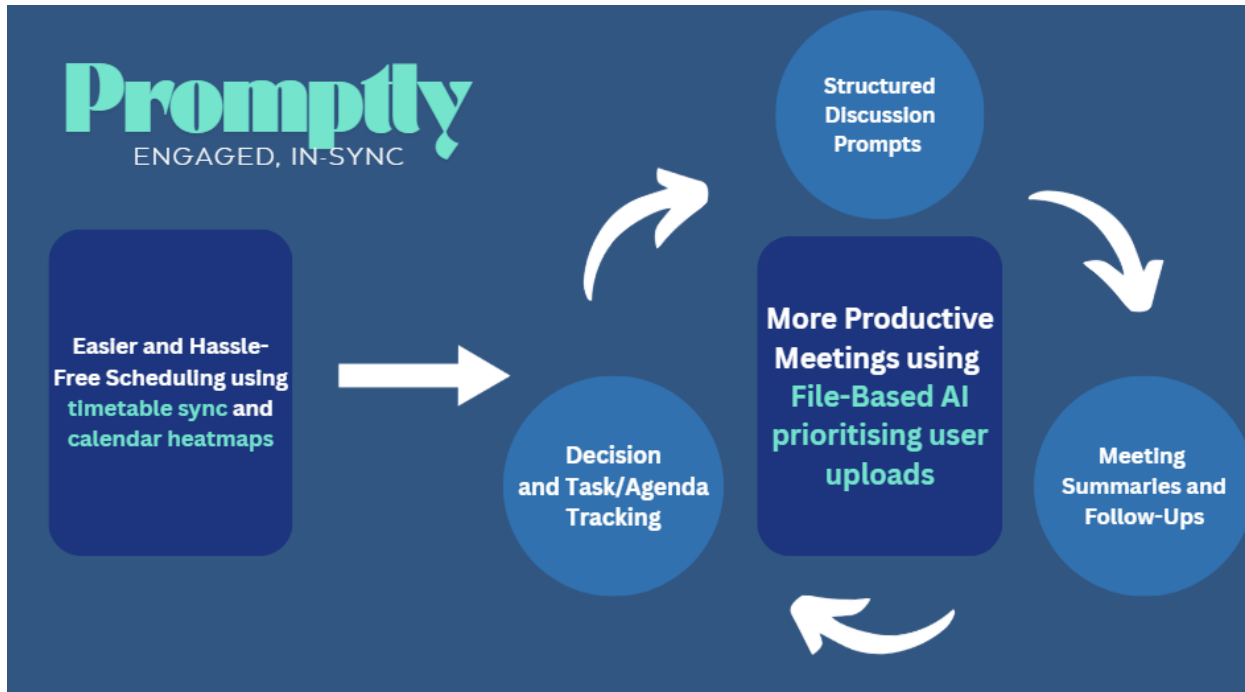


Fig 4.1: Flowchart Summary of Promptly User Requirements and Key Features

From the identified user needs such as more productive meetings and wants such as meeting structure and scribing, our team introduces Promptly: An Integrated meeting host system that streamlines the scheduling of online video conferencing and facilitates more productive meetings for university students.

Easier and hassle-free scheduling would be achieved through timetable sync and calendar heatmaps, while more productive meetings would make use of File-Based AI tools that prioritize user uploads.

4.1 Use of Artificial Intelligence (AI)

Our team thoroughly considered the use of AI in our design, where a key concern was accurate portrayal of information from AI generated prompts and suggestions, as well as keeping AI prompts within the context of the content in a group meeting. To tackle this, Promptly would make use of Retrieval Augmented Generation (RAG), which is an advancement in AI that focuses on analyzing User-Uploaded Files.

RAG incorporates an additional retrieval step before the generation process, allowing the AI to access relevant information, enhancing the accuracy of text generation (Arslan et al., 2024). An emphasis is placed on query and context, where RAG uses the user-input question to search an index created from uploaded files, then retrieves the most relevant section of the files as context.

4.2 Persona – Natalie

To meet user requirements and further communicate and express Promptly’s purpose and vision, our team introduces personas that characterize people who might use the product, highlighting their goals and frustrations.

Persona - The Organized Leader

Extroverted Conscientious Reliable Organised Friendly

Natalie is a straight-A student and a natural leader. She is the president of the on-campus K-pop dance club and an active member of the student business association. She loves all things matcha-flavored and her hobbies include travelling and hiking.

Natalie, 24

- Singaporean
- Year 4 Undergraduate
- Business & Communications Double Major
- Moderate Tech Proficiency

Goals

- Ensure **organization and efficiency** in group meetings
- To keep group members **motivated, engaged and accountable**
- A **quick and hassle-free** way to find free time among groupmates' schedules and **arrange group meetings**

Frustrations

- **Unresponsive groupmates** who take a long time to reply to messages to schedule meetings
- Groupmates **not actively participating** in the discussions
- **Awkward silences** in group meetings, forcing her to always be the one who speaks out to drive the meeting forward

Fig 4.2: Persona 1 (Natalie) - Photo from [Unsplash](#)

Firstly, we have Natalie, the Organized Leader. She is extroverted, friendly and organized as well as a Straight A student and natural leader. Natalie’s goals represent the requirements that Promptly aims to target, such as enhanced organization and efficiency as well as facilitating engaging group meetings.

Frustrations that were inspired from our primary research include unresponsive groupmates, inactive participation as well as awkward silences.

4.3 Persona – James

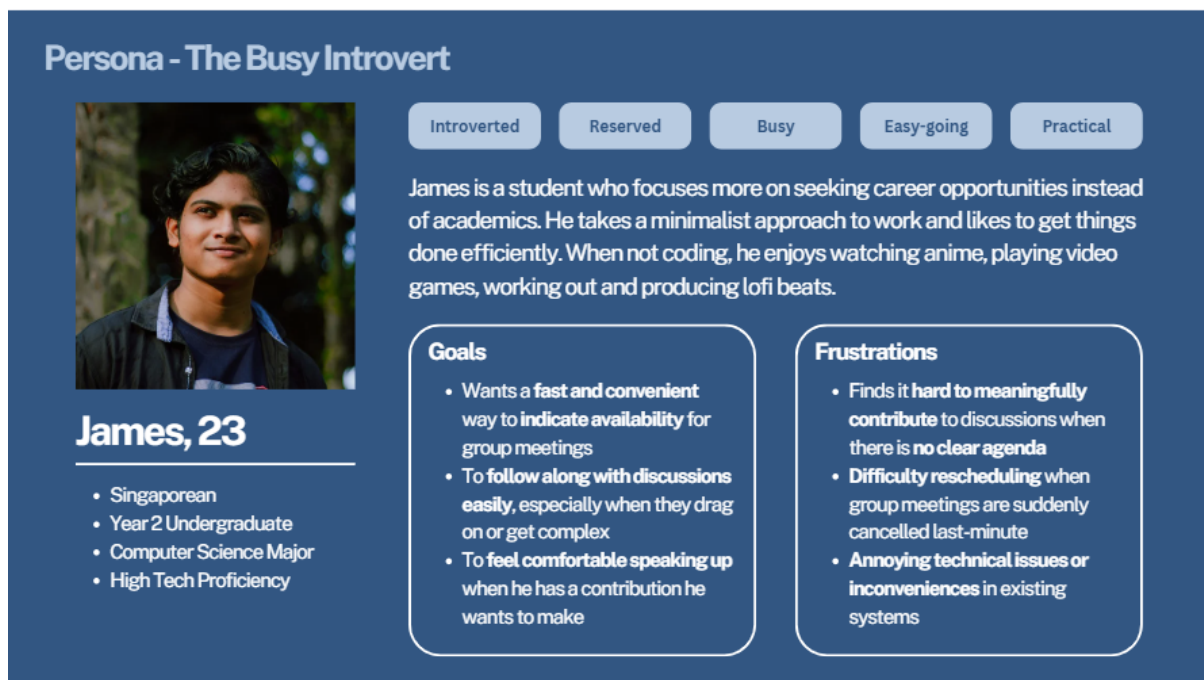


Fig 4.3: Persona 2 (James) - Photo from [Unsplash](#)

James is our second persona, The Busy Introvert. In contrast to Natalie. James is introverted, reserved and easy-going. He does not have a knack for academics, focusing more on developing his career. His main goals lie in convenience and communication, where he wishes to easily indicate his availability and gain confidence to speak up without being misunderstood.

Frustrations include annoying technical issues, difficulty in contributing meaningfully with no clear agenda as well as rescheduling difficulties.

4.4 Use Scenario (Natalie)

4.4.1 Current Use Scenario (Natalie)

Natalie is a final-year Business (Finance) major, currently working on her Field-Service Project (Final Year Project). Each student is expected to commit between 200 to 300 hours, which includes company visits (both individual and group), report writing, and presenting their findings to a real-world company's management.

As the assigned group leader, Natalie is working with four other members. They have completed the report and are scheduled to present their findings to the company in two weeks. However, with multiple deadlines and commitments from other modules piling up, the project has taken a backseat for most of the group.

Natalie checks Google Calendar and NUSMods to figure out her availability, only to realize her schedule is packed. Most of her evenings are filled with club activities, meetings, and other commitments. Still, she tries to carve out time wherever she can to focus on the project.

She then opens Telegram and sends a message to the group with a When2Meet link, asking everyone to fill it in asap. Three hours later, one member responds, but Natalie notices that they barely have any overlapping availability. A day later, another member fills it out. After three days, two members have not responded (MIA). Natalie tags them in the group chat to remind them. A day later, there was still no response. Frustrated, she sends them each a direct message.

After some back and forth on Telegram, the meeting finally happens but it is less productive than Natalie had hoped. One member cancels at the last minute, but the group decides to proceed anyway due to the looming deadline. Another member arrives 10 minutes late for the Microsoft Teams meeting.

When Natalie asks for input on how to structure the presentation, silence fills the call. She waits, hoping someone will speak up. Nothing. Awkwardly, she started outlining ideas herself in the Google Documents, with occasional hums of agreement and typing from the other 3 members. No one has their cameras on and Natalie wonders if they are scrolling through their phones or multitasking. It takes two hours for them to settle on an outline and assign tasks to each member.

4.4.2 Proposed Use Scenario (Natalie)

Natalie uses our interface: Promptly, to import her own calendar into the system. She sends a collaboration invite to her group members through Telegram and gets them to sync their calendars into the system. This required Natalie to input the purpose of the meeting, the estimated duration and the mode of meeting. Within an afternoon, all the group members clicked on the link invite sent and promptly synced their calendars together. Moving on, Promptly suggests a meeting time based on the cumulative availability of the group members: 26th March 2025 3-6pm over zoom. This is then sent as a calendar invite to the group members and requires them to accept or decline. Without many issues, the group members quickly accepted the invite, and the meeting has been noted on their respective calendars.

Fast forward to the meeting, Promptly sends out reminders to each group member 10 minutes before the meeting. This saw Natalie's group members to be on time, and the meeting started at 3:05pm. Natalie utilizes the Promptly's video call to conduct the meeting. As she got each group member to update on their individual progress, Scribby, the interface's AI scribe, starts to scribe for the meeting. As the meeting went on, Whispi, Promptly's chatbot, continued to suggest to members on what they could do/think of next. Whispi also calls out on group members that are inactive based on the status icon. The meeting ended in 1hr 10 mins and Scribby produced a summary for what has been discussed at the end of the meeting. This allowed Natalie and her group members to reference the ideas shared while they worked on their individual parts of the project.

4.5 Use Scenario (James)

4.5.1 Current Use Scenario (James)

James is a Year 3 Computer Science major. He is introverted, slightly reserved, and prefers spending his time on personal projects or part-time internships rather than group assignments. James has recently been assigned to a group for his Software Engineering module, which requires weekly group meetings to work on a project.

With little interest in leading or managing the group, James is mostly a passive participant. When his group starts discussing meeting times, James finds it bothersome to sift through chats in Telegram, where messages

pile up quickly due to off-topic jokes and late replies. The group leader shares a When2Meet link across the group chat, but it gets buried under a flood of subsequent messages about unrelated matters. By the time James notices it, it's already two days old, and he is unsure if everyone has submitted their availability.

The group finally agrees on a time that clashes with James's preplanned internship hours, forcing him to request a change. Rescheduling is tedious, and James is reluctant to ask again, feeling awkward to inconvenience the group. When he does make it to the meetings, he finds the agenda unclear. Often, the discussion stalls with lingering silences, or group mates talking over each other. James wants to contribute but hesitates to speak up unless prompted, fearing misinterpretation or redundancy. The meeting typically ends without clear action points or shared notes, making James unsure of his tasks till someone pings him afterwards on Telegram.

Overall, James leaves the meeting feeling disconnected and worries he did not contribute meaningfully. For him, the hassle of scheduling and the unstructured, ad-hoc nature of meetings reduces his motivation and confidence to participate.

4.5.2 Proposed Use Scenario (James)

James receives a link from his group leader to join the project workspace on Promptly. Skeptical at first, he is pleasantly surprised by the clean interface prompting him to link to his Google Calendar. With just a few clicks, he syncs his availability, and within minutes, Promptly automatically suggests optimal meeting slots for the group based on everyone's updated schedules, highlighting times that work for everyone. James does not need to trawl through chat history or polls, he simply accepts the proposed slot, and Promptly sends out a confirmation and automated calendar invite.

Before the meeting, James receives a clear agenda breakdown through the Promptly dashboard, showing discussion points and expected outcomes. At the start of the video call, Promptly's integrated AI scribe, Scribby, begins transcribing the discussion unobtrusively. James finds relief knowing he can review the transcript if he misses anything.

During the meeting, whenever discussion slows, the smart chatbot Whispi gently suggests relevant topics, new ideas, or prompts for input by calling on those who have been silent. James gets a private prompt suggesting, “Would you like to share thoughts on the frontend component?” This direct, non-intrusive nudge gives James confidence to speak up. He shares his input without fear of misunderstanding, as Whispi also offers concise AI-generated summaries and rewordings in real time.

After the meeting, the action items and key decisions are automatically summarized and assigned, visible in the project Meeting Pod. James can review what was discussed and his tasks anytime without waiting for someone to send notes. He feels included and less anxious about missing information or being overlooked, and with less friction in scheduling and discussion, James finds himself participating more actively and efficiently within the group.

Milestone 5: Conceptual Design & Prototype

Based on the functional and user requirements outlined in the previous section, we constructed a high-level conceptual design for our system, which subsequently guided the development of our preliminary concrete design in the form of a storyboard and a Miro lo-fi prototype.

5.1 Conceptual Design

Our interface is a virtual collaboration and video-conferencing platform that aims to seamlessly and efficiently integrate various aspects of productive meetings and workspaces into one shared platform. This includes providing users with a more streamlined and efficient method of scheduling meetings for projects, as well as a video-conferencing platform that improves productivity by reducing instances of awkward silences and inactivity. This interface also serves as a workspace whereby users can store their work online and utilize artificial intelligence agents to facilitate idea generation.

5.1.1 Overarching Conceptual Model: Meeting Pods

Our main interface metaphor will be office meeting pods, which are glass-enclosed soundproof booths that provide a conducive space for individuals to join virtual meetings, as well as small groups to conduct private in-person meetings. Considering our target audience of tertiary education students, we believe that adopting an office meeting pod design will create a more familiar and inviting interface, making it easier for students to engage with and embrace the platform. This ties in well with the rising popularity of meeting pods in office spaces, one which we believe tertiary students will resonate with when they enter the workforce. Using meeting pods as our overarching interface metaphor, we are able to model our interface based on the various features of a meeting pod. The system offers users their own virtual meeting pod, each denoted as one specific project.



Fig 5.1: Office Meeting Pods

As mentioned, the meeting pods can be used by users to carry out work for their individual projects. It is a compartmental organization whereby all things related to the specific project will be stored within that meeting pod. When users dive into a meeting pod, they will be greeted by a layout that resembles that of a workspace, with functional office objects that help them manage various aspects of the project. The workspace includes:

- A calendar that displays the user's schedule
- A filing cabinet that stores documents and files in a hierarchical file structure, including
 - User-uploaded files used in meetings
 - Meeting transcripts, summaries, and other files generated by Scribby & Whispi
- A contact list of the team members involved in the project
- A to-do list of tasks to be completed
- A meeting access point where users can join and host group meetings



Fig 5.2: Workspace within a Meeting Pod

5.1.2 Interface Metaphor: Calendar

Similar to the real world, calendars in our meeting pods allow users to view their project deadlines and schedule. Much like how calendars keep track of important events and deadlines, the calendar in meeting pods performs the same function. This calendar will display the various project deadlines and the user's overall schedule. With this task-domain object available across all meeting pods, it provides the user a quick shortcut to view their own schedule for planning and forecasting.

5.1.3 Interface Metaphor: Document Filing System

One of the key requirements for our system is for users to be able to upload files or documents related to each project, which provide contextual information and data to power the AI assistants that help to drive forward discussions during meetings. Moreover, users also require a way to easily access AI-generated transcripts and summaries for past meetings. To provide these features, we have decided to adopt a filing system within each meeting pod centered around the interface metaphor of physical folders and paper documents stored in filing cabinets, similar to what one would see in a physical office, combined with the organizational metaphor of a hierarchical tree structure that users are familiar with in other systems such as their computer desktop.

Each file or document is a task-domain object that stores textual or pictorial information and possesses attributes such as its file name, date added or modified, the user that uploaded it, and format. Similarly, folders are also task-domain objects that function like a container that stores references to other folders and files. Users can instruct the system to edit a folder or file's name, delete the file or folder and create a new file or folder. They can also manipulate the objects by dragging and dropping them into different folders for hierarchical organization, where different folders are nested within each other and each folder branches out into more folders and files.

Users can traverse through the folders to go further down the tree structure as well as backtrack to find their desired file. With the folder names serving as labels for users to categorize and group related files together, this filing system helps to reduce users' cognitive load by allowing users to customize the structure of their file storage according to what is easy for them to remember and promoting recognition over recall with similarly customizable folder and file names.

5.1.4 Interface Metaphor: Contact List

For each meeting pod, the system keeps a list of the group members involved in the project so that users within the same project can easily view the profiles and schedules of their group members as well as find ways to contact and communicate with them.

5.1.5 Interface Metaphor: To-do List

To help users more easily keep track of their project progress and plans following meetings and discussions, we intend to include a customizable to-do list of project tasks that users can access from their meeting pod. This component directly mirrors physical checklists or to-do lists that one might keep in their personal planner, where tasks that are to be completed are written down sequentially in a numbered list. This form of external cognition reduces memory load and aids them in remembering what needs to be done. When they complete a task, they cross it off or draw a line through it to denote completion, making use of annotation and cognitive tracing to track completed tasks so that they can ignore them.

In our interface's to-do list, project tasks are task-domain objects with attributes like task title or description, task completion status, assignee and relevant subtasks. Users can instruct the system to edit the title, delete or add a new task, mark the task as complete or incomplete, add an assignee, and add or delete a subtask. They can also manipulate the list itself by dragging and dropping tasks to reorder them, customizing the sequence in a way that makes sense to them.

5.1.6 Interface Metaphor: Virtual Meeting Rooms and Personal Assistants

Building on existing practices, our online video conferencing feature is designed as virtual meeting rooms. This seamlessly integrates with the overarching meeting pod metaphor, where all meetings within a pod take place in a virtual space and can be joined via a meeting access point. Moreover, allowing users to communicate through both video and audio helps to promote telepresence, where users can see and respond to each other in real time despite physical separation, thus allowing discussions to take place more efficiently and effectively. We aim to increase productivity further by introducing activity indicators for each participant, in the form of on-screen status symbols that take on a different appearance depending on the user's level of participation. This makes use of social translucence to encourage group members to participate in the discussion more, as making their activity level visible to others in the group increases social pressure to conform and contribute to the overall discussion.

Building on the office space and meeting pod concept, we introduce AI-driven meeting assistants, Scribby and Whispi, designed to act as personalized virtual assistants which enhance user interaction and engagement. Scribby acts as an intelligent scribe and note-taking assistant that transcribes discussions in real time as well as provides summary suggestions, catching and organizing key pointers during the meeting. To interact with Scribby, users can instruct the system to expand or hide the live transcript and meeting summary generated, export or download their content as documents, and enable editing of the meeting summary.

On the other hand, Whispi facilitates discussion and serves to keep meetings productive by providing contextual assistance based on user-uploaded files and contributions made by users during the meeting itself. Users can converse with the Whispi chatbot via text like how they would interact with an additional group member

present remotely without audio or video enabled. This way of communication is also used in popular AI-driven chatbots already on the market, and with many students already being familiar with these existing systems, they will find engaging with Whispi to be not only intuitive, but also seamless and convenient given its integration into the meeting environment itself.

5.2 Concrete Design

With our conceptual design in place, we made a preliminary concrete design to better illustrate our system's features and user experience while providing materials for initial user testing.

5.2.1 Storyboard

We produced a storyboard from the perspective of one of our personas, Natalie, based on the scenario outlined in our requirements definition section. This helped us to further develop and concretize aspects of the user experience throughout the whole process, from scheduling a meeting, then during the meeting itself, to finally leaving and following up after the meeting.

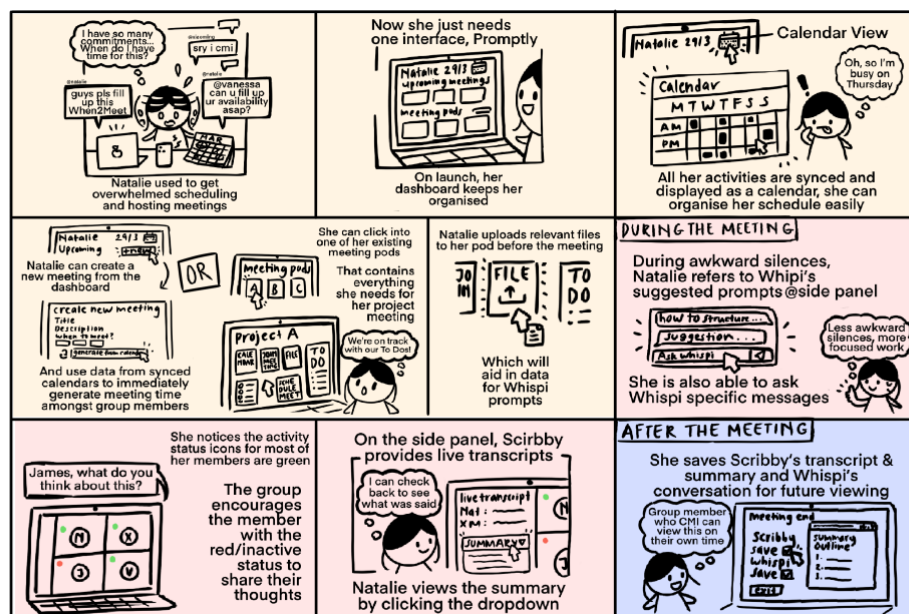


Fig 5.3: Storyboard depicting Natalie using Promptly to schedule and facilitate group meetings

5.2.2 Lo-fi Prototype

To provide an interactive prototype for testing, we created a low fidelity prototype on Miro, which includes a wireframe of the general screens and interaction flow involved in using the system. While it does not possess actual interactivity, it makes use of arrows to show the relationships between components that will aid us in the Wizard-of-Oz technique during user testing. Below are brief descriptions of major features and design decisions made in the prototype. The Miro board containing the entire prototype can be found [here](#).

Dashboard

When users first log into the system, they are greeted with a dashboard that provides an overview of featured meeting pods and upcoming meetings. This overview ensures the interface maintains its minimalist design while also taking advantage of the idea of external cognition to reduce user's cognitive workload by showing important details from the get-go. This prevents users from having to navigate deep into the system to look for such important details.

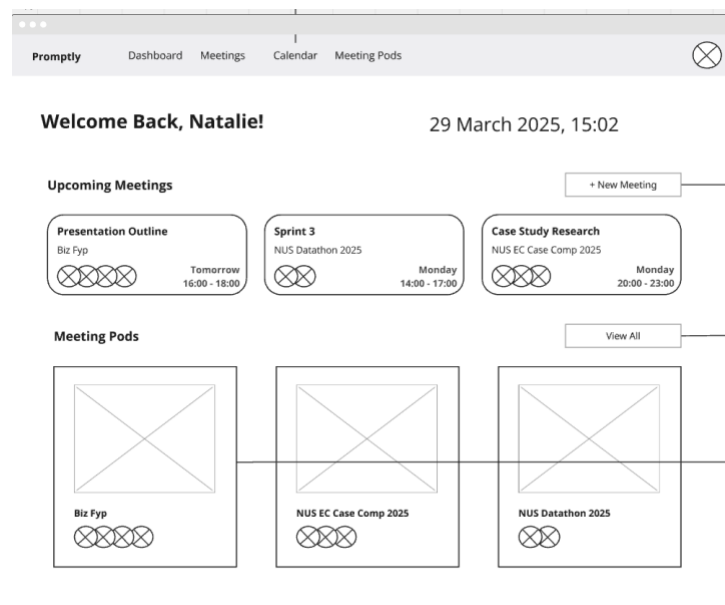


Fig 5.4: Dashboard or Home Page

Meeting Pod

From the dashboard, users can click on the ‘View All’ button to view all the meeting pods for the projects they are currently a part of or working on.

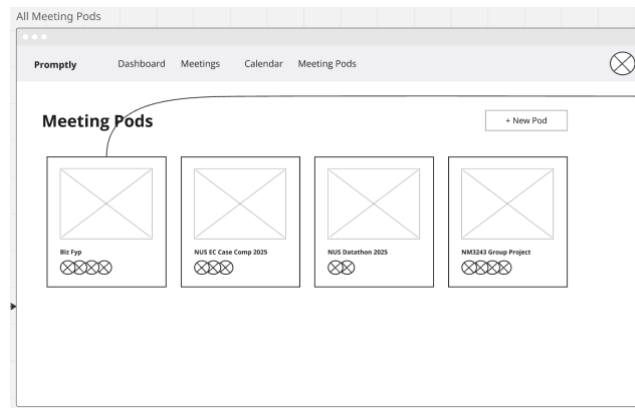


Fig 5.5: All Meeting Pods

From this page, they can click on each meeting pod, which will bring them to the home page of that meeting pod. Here, they can view all the information related to the project and access project-specific features such as joining upcoming meetings, accessing project documents and viewing the project to-do list.

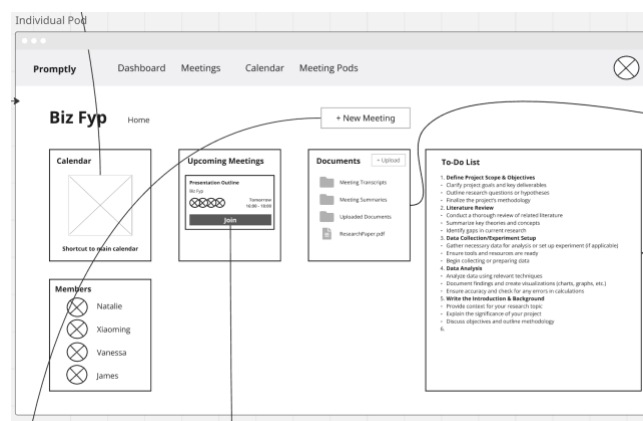


Fig 5.6: Individual Meeting Pod Home

Calendar

The user's calendar can be accessed via the dashboard, top bar or the individual meeting pod home page. While a typical calendar may be displayed by month, we decided that our interface's calendar needs to be displayed by week. This allows users to have a more detailed breakdown of their own schedule, down to the hour. Additionally, leveraging on the notion that vision is the most dominated sense for sighted individuals, we decided to color code the various events by projects and purpose. This reduces the cognitive workload of the user by making information more immediately perceptible.

When the user chooses to generate a common available date among the meeting participants when scheduling a meeting, they will be brought to the Calendar view with a heatmap that indicates possible time slots to schedule the meeting. The darker the shade of green, the more participants are available during the time slot. The use of contrast directs their attention to the darker slots, allowing them to quickly find the most optimal slots. Moreover, the simplistic visual representation gives users just enough information to make an informed choice while not overloading them with distracting UI elements that can impair their decision-making.

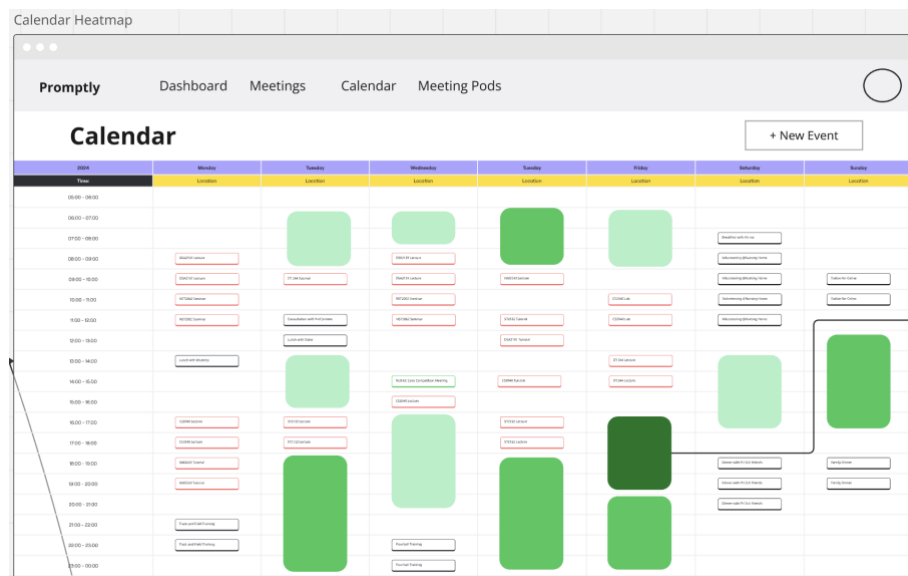


Fig 5.7: Calendar with Heatmap

Meeting

During a meeting, users are greeted with features that provide both the basic and enhanced functionality required for seamless remote meetings, all organized in a familiar layout modelled after current systems on the market that makes it easier for students to navigate and adapt to the interface. This includes:

- User Grid Display: A structured view of all participants in the meeting.
- Bottom Toolbar: A centralized control panel for key functions.
- Mute & Video Controls: Positioned on the left side of the toolbar for quick access.
- Status Icon: Indicates user engagement and accountability within the virtual space.

Both AI meeting assistants are designed to be helpful yet unintrusive to prevent information overload and annoying the user, thus requiring the user to click on a button in the bottom toolbar to make each of the assistants visible. If the user decides to show the AI assistants, they will appear stacked on the left side of the screen, allowing users to have an easy view of everything even if both assistants are enabled.

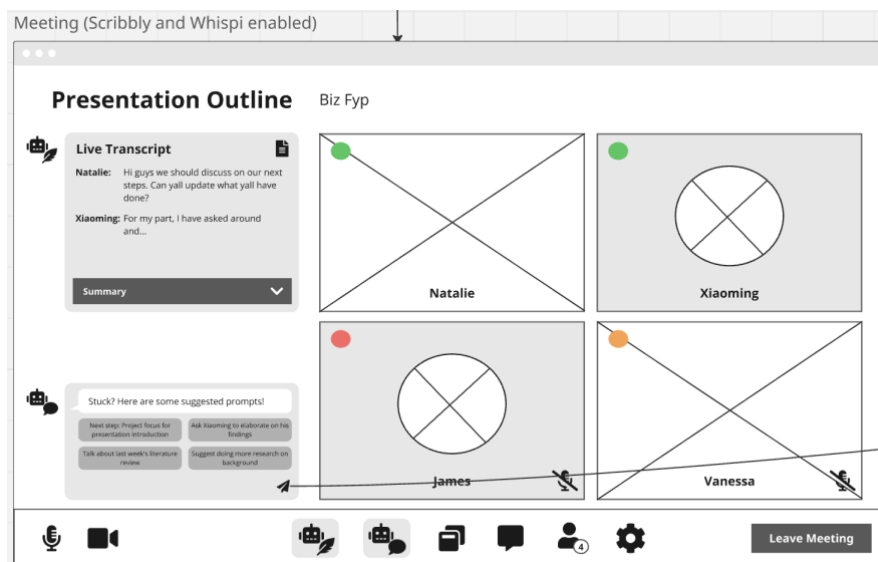


Fig 5.8: Meeting

Milestone 6: Evaluation with users

With our preliminary lo-fi prototype, we conducted a formative evaluation with users to determine whether our design meets users' needs and discover ways to improve on it further.

6.1 User Testing Plan

We decided to conduct remote usability testing with 4 users from our target audience, making use of our lo-fi Miro prototype and simulating interactions by dragging the screen to show the result of the actions the user intends to perform. Users are made to think aloud and vocalize their perceptions, opinions, frustrations and general thought process using our system, so that we can not only observe how they react and respond to our system, but also uncover gaps or errors in their mental models of how our system works. We created a testing script with guidelines for the study and interview questions to ask the user, which can be found in Appendix C.

The general structure of the usability test is as follows:

1. Pre-interview brief and informed consent form
2. Users instructed to perform a series of 4 tasks corresponding to user tasks derived from user requirements:
 - a. Viewing and accessing calendar, upcoming meetings and meeting pods
 - b. Creating and scheduling a new meeting by syncing calendars
 - c. Joining and participating in a meeting supported by AI assistants, Scribby and Whispi
 - d. Accessing files and documents related to each project, e.g. past meeting transcripts and summaries
3. Post-test interview questions (qualitative and quantitative) and debrief

6.2 User Test Findings

From our usability tests and interviews, we have gathered valuable feedback and insights on how to improve our design. In general, our users found our system's interface quite intuitive due to its resemblance to other video conferencing apps they are familiar with. They mostly found it straightforward to complete desired

goals in our system and that the system's features were helpful for them in scheduling and hosting group meetings. Some also appreciated the minimalist and clean design for certain screens, such as the dashboard and meeting view. Meanwhile, we were also able to uncover problems with our system that impacted user experience. Some of the more major problems are outlined below, with the corresponding proposed redesigns detailed under Milestone 7.

6.2.1 Excessive Calendars

Some users were confused about whether the 'Calendar' shortcut in the individual meeting pod home page directed them to the same calendar as the 'Calendar' button on the top bar of the screen, or if it was for a different calendar. Both were intended to be linked to the user's personal calendar with their entire schedule, but it is understandably confusing when we provide two different ways to access the same calendar in the same screen, with one of them given in the context of the meeting pod. This reveals a mismatch between the user's mental model of how our system is organized and what we intended.

6.2.2 Meeting Pod Layout

Some users found the meeting pod layout to be too rigid. They wanted the ability to organize it based on their team's workflow (e.g. by adding important shortcuts like Google Drive or GitHub links, or having separate areas for notes or discussion points beyond the to-do list). Another user had trouble locating the 'Join Meeting' button and felt that if they were rushing to join a meeting, they should be more easily be able to locate a meeting in progress.

6.2.3 Dashboard Shortcuts

The prototype did not offer navigation straight to a meeting pod or an upcoming meeting despite them being prominently displayed on the dashboard. User was confused why they had to click 'View All' then click the meeting pod when they could have clicked on the dashboard module straight away.

6.2.4 Calendar Heatmap

There was ambiguity as to the number and identities of groupmates available for each suggested slot highlighted in the calendar heatmap when the user tries to generate an overlapping time slot from groupmates' calendars as the system did not explain which levels the different shades of green correspond to.

6.2.5 Meeting Timer

Given that the meeting is scheduled for a certain duration, there is no way to track the time elapsed during the meeting, making it easy for the group to lose track of time and be unproductive as a result.

6.2.6 Whispi Interface

Whispi's suggested prompts for the user to use in the verbal discussion to drive it forward were mistaken for suggested inputs into Whispi's chatbot conversation. A user thought they were buttons to click on, and that the paper airplane icon meant sending the prompts as inputs all at once, instead of a button that expands the text field for user input. Moreover, some users noted that when both Scribby and Whispi were enabled, the screen appeared rather cluttered.



Fig 6.1: Issues with the Whispi Interface

6.2.7 Activity Indicators

Users were confused as to what the colored dots (activity indicators) meant, and whether they are interactable or serve other functions beyond indicating a groupmate's current level of participation.

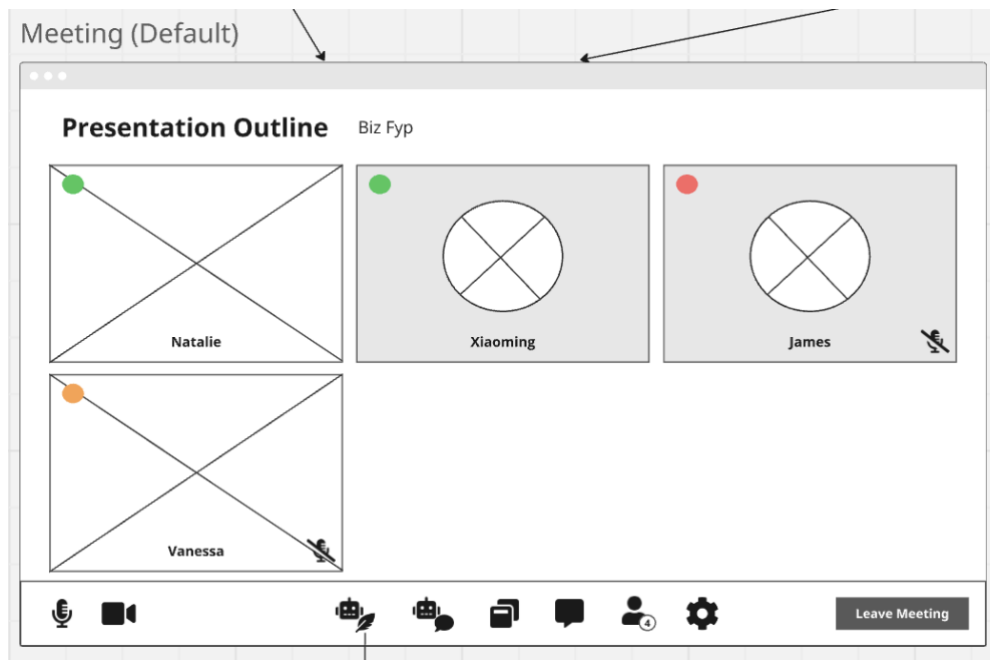


Fig 6.2: Colored Activity Indicators

Milestone 7: Modifications

As we evaluated the feedback from our evaluation with users, we came up with a few modifications to our prototype and overall design of our project. These changes were made with design considerations in mind.

7.1 Revised Prototype

The Miro board containing our revised prototype can be found [here](#).

7.1.1 Calendar

We modified the interface and overall conceptual model to distinguish between global and local structure. This meant that the Calendar in the menu bar, ordered across the top row of the interface, refers to the user's global calendar, which consists of the user's entire timetable and calendar with all their deadlines and scheduled meetings. On the other hand, the calendar in each meeting pod will display a calendar that is filtered to only show upcoming deadlines and meetings for the specific project. This allows the user a much clearer view of the project progress and requirements. This is further emphasised by our hierarchical organization of the interface whereby there is a clear workflow in using the interface.

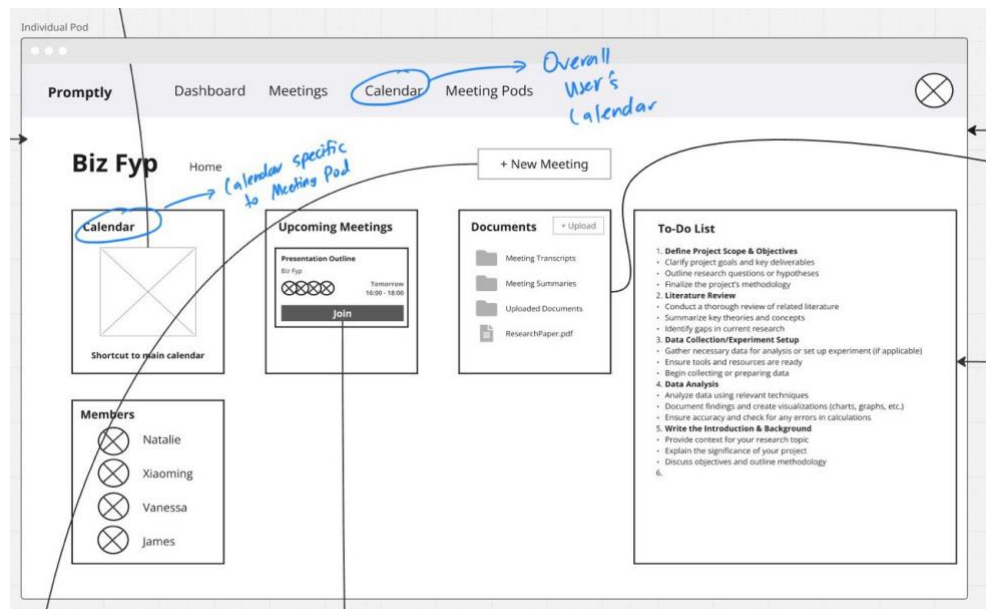


Fig 7.1: Calendar in Interface

7.1.2 Revised Meeting Pods

We introduced more customization with modular sections that users can add, reorder or remove together with custom labels and icons to tackle the problem of rigidity of meeting pods. This improves user experience in terms of the interface's cognitive aspect. More specifically, this feature accounts for each user's attention in terms of the way in which information is being presented. With our interface aimed at improving productivity, ensuring that the interface is tailored to user's preferences will help in improving their attention span and hence result in higher productivity.

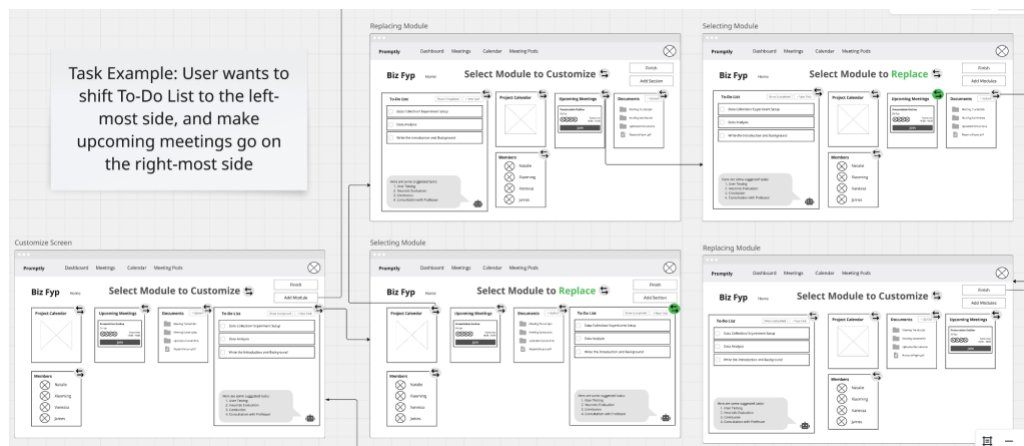


Fig 7.2: Modular Meeting Pod

7.1.3 Dashboard Shortcuts

As the initial design did not offer navigation straight to a meeting pod or upcoming meeting, despite the illusion of such, we will introduce shortcuts to bridge the gulf of execution. This will allow the interface's conceptual model to be more closely aligned with the user's mental model.

7.1.4 Revised Calendar Heatmap

To tackle the ambiguity in the number and identities of groupmates available for each suggested slot, the interface will provide details such as the number of group members available in each time slot and who exactly is available when the user hovers over the slot. This will help in improving visibility of system status.

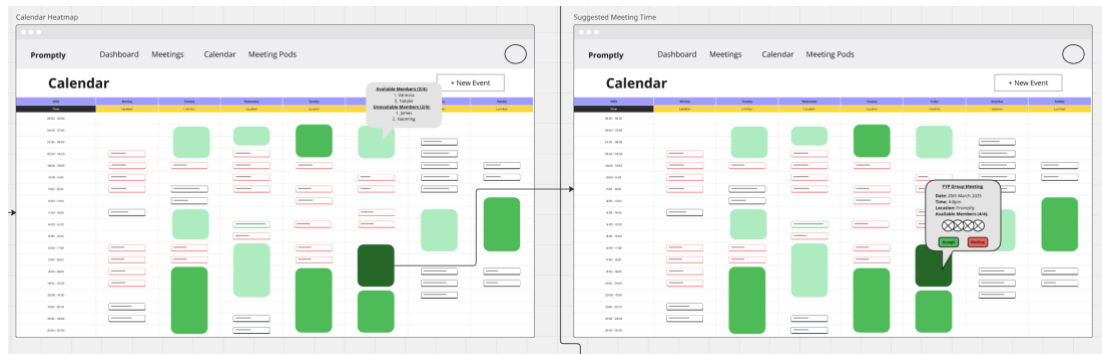


Fig 7.3: Revised Calendar Heatmap

7.1.5 Meeting Timer

With the objective of our interface to improve productivity, we will include a meeting timer to allow users to be accountable as they work towards completing their meetings under the scheduled time.

7.1.6 Whispi Interface

Users identified that the Whispi chat and the meeting chat clutters the screen, making it confusing to utilise both Whispi and the meeting chat. As such, we reduced Whispi into a popup that utilises the same chat segment as the meeting chat. Upon activation, the meeting chat will be overlayed by Whispi and users can ask questions.

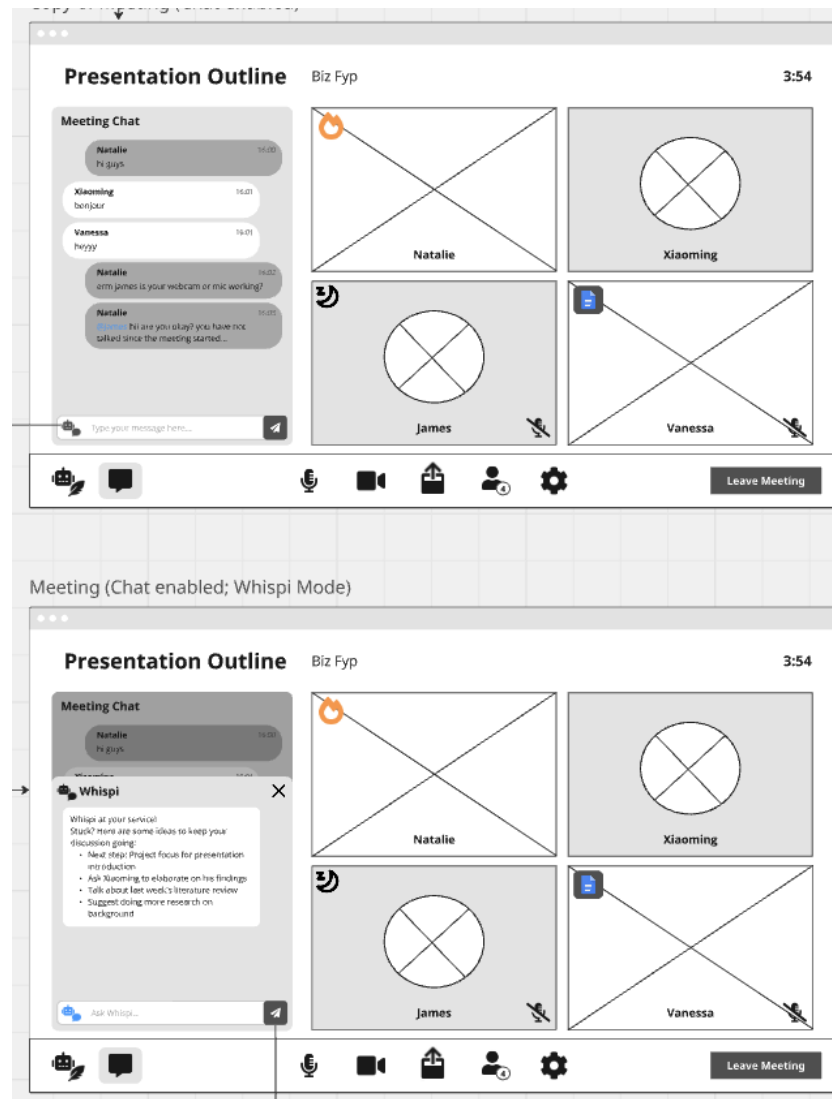


Fig 7.4: Revised Whispi Interface

7.1.7 Activity Indicators

To ensure that the interface is accessible to people with colour-blindness, as well as users highlighting their inability to understand what the activity indicators are, we will be introducing 3 activity indicators, together with a neutral state with no activity indicator:

1. Flame: The user is 'on fire' and actively leading the discussion
2. None: The user actively participates but only sometimes
3. Moon: The user is 'asleep' and not active in the discussion
4. Custom: The user is not active in the discussion but is working on the project elsewhere or offline

By default, the system tracks the user's level of participation in the verbal discussion as well as the meeting chat and updates the activity indicator accordingly (i.e. as either flame, none or moon). The introduction of the custom status is to prevent misunderstandings in cases where a user is thought to be slacking off due to their 'asleep' status as a result of them not speaking much or not being active in the chat, even though they are still meaningfully contributing elsewhere (e.g. working on the project report on Google Docs). In such cases, the user can manually opt out of activity tracking by setting a custom status to notify the group that they are still contributing, though not verbally. Group members are then to manually check on this user's progress on the indicated platform to make sure that they are still active.

Altogether, these activity indicators are designed keep the participants accountable during the meeting to discourage being disengaged and encourage productivity.

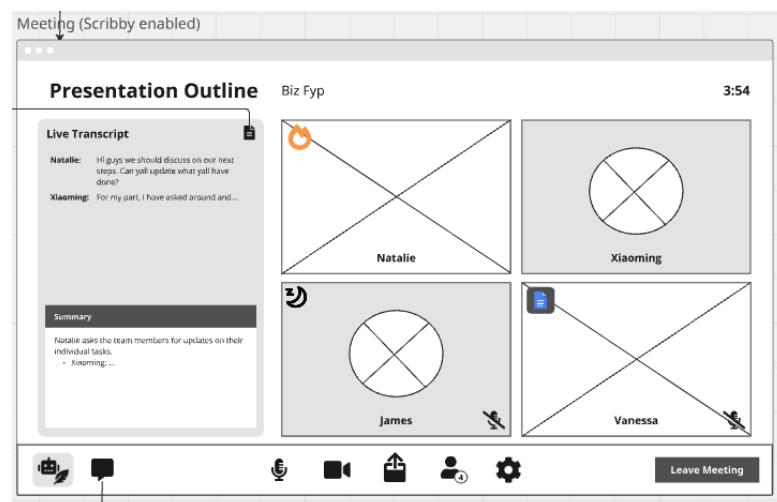
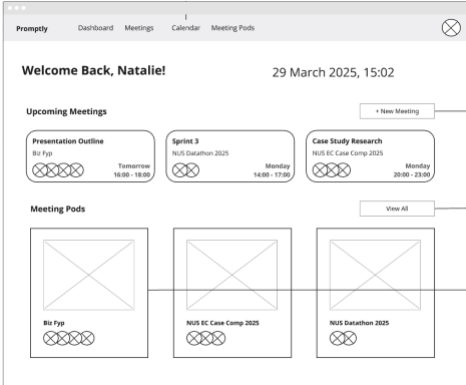


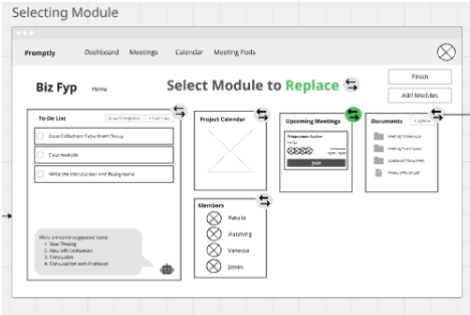
Fig 7.5: Revised Activity Indicators

7.2 Expert Testing & Heuristic Evaluation

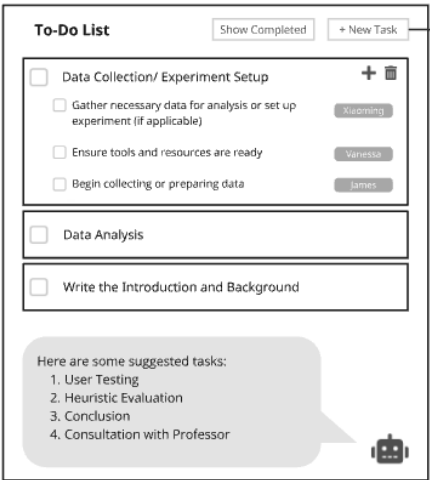
Following the user evaluation process and our modifications to our interface, we subjected ourselves to a heuristic evaluation to ensure product usability. Each member evaluated the redesigned interface with Jacob Nielsen's 10 usability heuristics. A brief description of the issue was provided, followed by a reference to the issue, the heuristics violated, an objective rating of the issue and a proposed solution, as shown in the table below.

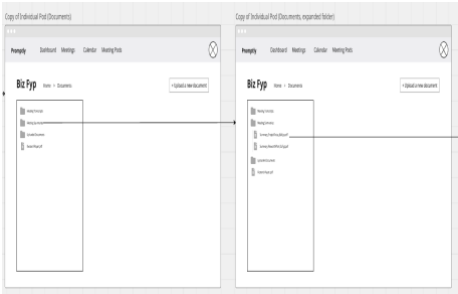
No.	Brief Description of Issue	Heuristic Violated	Ratings	Proposed Solution
1	The Upcoming Meetings on the Dashboard suggests the ability to click on them as a shortcut to the meeting call. However, it is merely a reminder. 	2: Match between System and the Real World 4: Consistency and Standards 6: Recognition Rather than Recall	How common: 1 Hard to Overcome: 1 On-off or Persistent: 1 Perceived Seriousness: 1 Overall: 1	Redesign to a more intuitive design such that it does not make users feel like it is a shortcut to the meeting room/call.

2	When editing the meeting pods together as a group, no warning and undo if one member were to delete/edit everything.	5: Error Prevention	How common: 2 Hard to Overcome: 1 On-off or Persistent: 3 Perceived Seriousness: 3 Overall: 3	Add a warning popup when a user is about to make a major edit/deletion, have an undo button/history to recover past items.
3	No tutorial or guide/manual for new users, no existing help/button for when users are unsure what to do.	6: Recognition Rather than Recall 10: Help and Documentation	How common: 2 Hard to Overcome: 1	Create a button for users to get help when they are stuck that will introduce a pop up with information on how to proceed.

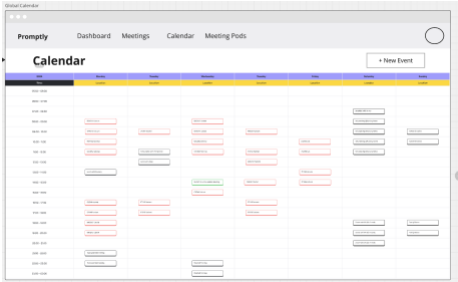
			On-off or Persistent: 2 Perceived Seriousness: 2 Overall: 2	
4	The Message in the ‘Customize Meeting Pod’ Page intends to instruct the user to select the ‘shuffle’ icon in the corner of individual modules. The user may misinterpret this instruction by attempting to select the module box instead of the corner button. 	2: Match between System and the Real World	How common: 2 Hard to Overcome: 1 On-off or Persistent: 2 Perceived Seriousness:	Allow the user to select either the icon or the module when attempting to customize their meeting pod.

			2 Overall: 2	
5	Users might not know how to edit the todo-list tasks due to the ‘hidden’ interactions and operations to edit the tasks and move them around. These hidden interactions are not found elsewhere in the system (users typically need to click on an ‘edit’ button to enter edit mode).	4: Consistency and Standards	How common: 1 Hard to Overcome: 1 On-off or Persistent: 1 Perceived Seriousness: 2 Overall: 2	Either add an edit mode where users are free to edit the todo-list, which will be saved after exiting the mode, or provide a help tooltip that tells users how to interact with and edit the todo-list upon hover. This can also be solved with a tutorial introduction for new users.

6	Users might accidentally edit the todo-list while navigating the meeting pod and clicking around (as there is no ‘edit’ button and they can freely edit anytime upon a double-click). 	5: Error Prevention 9: Help Users Recognize, Diagnose, and Recover from Errors	How common: 1 Hard to Overcome: 2 On-off or Persistent: 3 Perceived Seriousness: 3 Overall: 3	Provide an explicit edit mode for each current task on the todo-list where they have to explicitly exit from to save the edits (click on a tick or confirm button to save changes instead of clicking off the task).
7	There might also be conflicts where different users are editing the same task at the same time and there is no way to tell when the todo-list is currently being edited by someone else.	1: Visibility of System Status 3: User Control and Freedom	How common: 1 Hard to Overcome:	When a group member is currently editing a task, the edit button for that particular task could be greyed out to prevent others from editing at

			2 On-off or Persistent: 3 Perceived Seriousness: 4 Overall: 3	the same time and cause conflicting changes.
8	Users navigating the document section of the meeting pod might be lost in terms how backward navigation. While we provide a small navigation indicator beside the meeting pod title, beginners might not be aware of it to use it as backward navigation. 	1: Visibility of System Status 3: User Control and Freedom	How common: 1 Hard to Overcome: 3 On-off or Persistent: 3	We could include backward navigation buttons such as 'back' on the top of the interface to assist users in performing exits on our interface.

			Perceived Seriousness: 3 Overall: 3	
9	Beginner users may not be aware of the activity indicator and may be uncomfortable with it. They should be given a choice in allowing these features as the current modality only allow the host to dictate such preferences	1: Visibility of System Status	How common: 2 Hard to Overcome: 1 On-off or Persistent: 3 Perceived Seriousness: 3 Overall: 3	We should allow users to choose if they would like activity status to be active for them or not before joining a meeting. This speaks to privacy and informed consent concerns.

10	Users may assume that their global calendar has to be manually processed into the interface. This is due to the lack of features that allow users to upload calendar files downloaded from various calendar applications. 	1: Visibility of System Status 6: Recognition Rather than Recall	How common: 2 Hard to Overcome: 3 On-off or Persistent: 3 Perceived Seriousness: 3 Overall: 3	We should include a feature beside 'New Event' to allow users to upload calendar files downloaded from other calendar applications such as Google Calendar.
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Milestone 8: Conclusion

Overall, our video-conferencing and scheduling interface aimed at solving 2 main issues, namely a more efficient and hassle-free way of scheduling meetings with group members, and improved productivity during meetings in terms of greater structure in meetings, reducing inactivity of members during discussions and stir conversation.

In order to design an efficient and hassle-free way of scheduling meetings with group members, our interface utilizes calendar heatmaps produced from shared data of group members' calendars. Running the data through an optimized AI agent will be able to allow users to find the optimal time to meet with ease.

With regards to improving productivity in meetings, our interface strives to do so by utilizing AI assistants such as Scribby to help users scribe and summarize the meetings. Similarly, Whispi serves as an AI assistant that provides suggestions on how the discussion can flow based on the data collected within the meeting pod. Lastly, we utilized simple activity indicators as a means to keep users accountable during the meeting, where being inactive will be highlighted.

Even so, our interface failed to account for the offline implementation of our system. In other words, how does our interface adapt in an offline meeting? What are the specific key requirements and problems that arise in offline meetings? Needless to say, there is potential for the interface to develop to consider offline meetings requirements and how the interface could potentially utilize certain existing features, or introduce new features, to do so.

In conclusion, our interface can be seen as one that aims to provide efficient and hassle-free scheduling of meetings within group members and improved productivity of meetings and discussions through the use of A.I. assistants and agents. In an increasingly digital age shaped by rapid technological advancement, we believe our interface, grounded in thoughtful and user-centric design, effectively addresses the core challenges we initially set out to tackle.

Appendix A: Semi-Structured Interview Format

Pre-Interview (Consent Form)

1. Voluntary Participation: Your participation in this study is entirely voluntary. You may decline to participate or withdraw from the study at any time without facing any consequences.
2. Confidentiality: All information provided will be kept confidential and will only be used for the purposes of this research. Any personal details or responses will be anonymized in the final findings.
3. Voice Recording: Please be informed that a voice recording will be made during the interview. This recording is solely for the researcher's use in data analysis and transcription and will not be made publicly available.
4. Context: Hello! We are a group of NUS students conducting user research for our NM3243 User Experience Design project. Our goal is to understand the experiences of tertiary students with scheduling and hosting group meetings for collaborative work. The insights gathered will help us design a system that streamlines the scheduling process and enhances meeting productivity.
5. Demographics: Which year are you in? Which faculty are you from?

General Questions

1. How often do you participate in group meetings?
2. What are your group meetings usually for? (e.g., class projects, presentations, extracurricular activities)
3. Do you enjoy having group meetings? Why or why not?
4. How many people are usually involved in the group meetings you've participated in?

Scheduling Meetings

1. What do you use to keep track of your class timetable and personal schedule?
 - a. Probe: What do you like and dislike about these tools/systems? Why?
2. Can you describe the process of scheduling a meeting with your groupmates from start to finish?

- a. Probe: What tools/websites/apps do you use in the process?
 - b. Probe: How long does it take to finalize a date and time?
3. What do you like and dislike about the scheduling process? Why?
4. Which parts of the scheduling process do you think could be improved, and how?
5. When a group member can't attend a meeting, what typically happens?
 - a. Probe: Does this change depending on whether the notice is given last-minute or in advance?
 - b. Probe: How does this make you feel?
6. How often do you have to reschedule meetings?
 - a. Probe: What usually happens when a meeting is rescheduled?
 - b. Probe: How do you feel about rescheduling meetings? (You may say you don't feel anything)

Hosting Meetings

1. What meeting host/video conferencing software do you usually use for your meetings?
2. What is your preferred software for group meetings? Why?
 - a. Probe: What do you like or dislike about this software?
 - b. Probe: What about other apps or software?
3. Can you describe a typical group meeting from start to finish?
 - a. Probe: Do you or your groupmates typically use screen sharing during meetings?
4. What do you like or dislike about the screen-sharing feature?
 - a. Probe: Do you use external tools for collaboration during the meeting? If so, what are they?
 - b. Probe: What do you like or dislike about using these tools during meetings?
5. How long does a typical meeting last?
 - a. Probe: What do you consider to be the ideal meeting duration? Why?
6. How much of the meeting is usually spent on productive work?
 - a. Probe: What usually happens between productive tasks?
 - b. Probe: What problems affect the productivity of your group meetings?

7. How do you keep track of points raised during meetings for reference or follow-up?
 - a. Probe: Is there a dedicated person responsible for taking meeting minutes? If not, why?
 - b. Probe: Do you like this method of note-taking? Why or why not?
8. What do you like or dislike about attending group meetings? Why?
9. Which aspects of the group meeting experience do you think could be improved, and how?

Proposed Solution

Context: Imagine a company develops a new system designed to streamline meeting scheduling and improve productivity. The system would include:

- Automated schedule synchronization to find optimal meeting times,
 - Seamless transitions between external collaborative tools,
 - AI-assisted tools for note-taking and driving discussions forward.
1. How willing would you be to use this new system? Why or why not?
 - a. Probe: Which features of this system do you find appealing? Why?
 - b. Probe: Which features do you find unnecessary or undesirable? Why?
 - c. Probe: What additional features or functionalities would make you more likely to try this system?
 2. What features would you like to see in this new system? Please feel free to recommend any improvements.

Debrief

This is the end of the interview. Thank you for your time and feel free to contact us if you have any other questions.

Appendix B: Full Online Questionnaire via Google Forms.

General Details

This section asks you general questions about your experiences with project group meetings. Group meetings can be either physical (where members meet in person) or remote (where members meet online using video conferencing software).

1. Which year are you in? *

- Year 1
- Year 2
- Year 3
- Year 4+

2. What faculty are you from? *

- Faculty of Arts and Social Sciences
- Faculty of Engineering
- Faculty of Science
- Other: _____

Scheduling and Hosting Group Meetings

This section focuses on your experiences scheduling and hosting group meetings.

3. How often do you participate in group meetings (both physical and remote)? *

- Never
- Rarely (a few times a semester)
- Sometimes (a few times a month)
- Often (a few times a week)
- Always (every day)

4. For what function do you most often have group meetings? (Select all that apply) *

- Internships/Part-Time Work
- Class Projects
- Extracurricular Activities
- Other: _____

5. On average, how many participants are there in each group meeting you've attended? *

- 1-3
- 4-6
- 7-10
- >10

Scheduling Meetings

This section asks about your experiences scheduling and arranging meeting times.

6. Which of the following tools do you use to keep track of your timetable or schedule? (Select all that apply) *

- NUSMods
- NTU STARS Planner
- Google Calendar
- Outlook Calendar
- Apple Calendar
- Physical Journal or Planner
- Other: _____

7. Which communication channels do you usually use to discuss meeting times with groupmates? (Select all that apply) *

- SMS
- Email
- WhatsApp
- Telegram
- Discord
- Microsoft Teams
- Other: _____

8. How do you usually decide on a date and time to meet up with your groupmates? (Select all that apply) *

- Chatting in group chats
- In-person discussions
- Polling in group chats
- Polling on online platforms/apps (e.g. Doodle Poll)
- Using external tools to find overlapping free time (e.g. When2Meet)

- Sending invites via calendar apps
9. How long does it usually take to decide on a time to meet? *
- A day or less
 - A few days
 - A week
 - More than a week
10. To what extent do you agree with the following statement: "I find the process of arranging a time to meet with my groupmates to be efficient." *
- Strongly Disagree
 - Disagree
 - Neutral
 - Agree
 - Strongly Agree
11. What problems do you usually encounter when scheduling meetings with groupmates? (Select all that apply) *
- Groupmates taking too long to reply
 - Tedious to find common free time in everyone's schedule
 - Constantly having to reschedule meetings
 - None of the above
 - Other: _____

Hosting Meetings

This section asks about your experiences hosting and participating in group meetings.

12. On a scale of 1 to 5, to what extent do you find yourself attending remote meetings compared to physical meetings? (1 for only physical meetings, 5 for only remote meetings) *

- 1
- 2
- 3
- 4
- 5

13. What types of meetings do you find yourself partaking in most often? (Select all that apply) *

- Collaborative Work (e.g., working on a report/designing slides together in real-time)
- Ideation / Brainstorming Sessions
- Planning / Discussion Sessions
- Status Update / Check-in meetings
- Other: _____

14. What external tools do you usually use for collaborative tasks during group meetings? (Select all that apply) *

- Online document editors (e.g., Microsoft Word, Google Docs)
- Online slide deck editors (e.g., Google Slides, Canva)
- Digital whiteboards (e.g., Miro, FigJam)
- Technical tools and software (e.g., Coding IDEs, Painting Software, Microsoft Excel)
- Other: _____

15. How long does each meeting usually last? *

- < 0.5 hours
- 0.5 - 1 hour
- 1 - 2 hours
- 2 - 3 hours
- 3 - 4 hours
- > 4 hours

16. What problems do you usually encounter when attending and participating in group meetings? (Select all that apply) *

- Awkward silences
- Discussion going off-topic
- Difficult to keep track of points raised
- Lack of meeting minutes
- Not knowing who is talking during the discussion
- Disagreements or conflicts that are hard to overcome

- None of the above
- Other: _____

17. To what extent do you agree with the following statement: "I am satisfied with how productive my group meetings usually are." *

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

Making Group Meetings More Hassle-Free and Productive

This section asks for your opinions on a proposed solution aimed at streamlining scheduling and facilitating more productive discussions.

18. How willing would you be to use an all-in-one meeting host (or video conferencing) website/app that:

- Helps to automatically sync schedules to find optimal meeting times
- Allows for seamless transitions between external tools
- Provides AI-assisted tools that automate note-taking and drive discussions forward
- Very unwilling
- Somewhat unwilling
- Neutral
- Somewhat willing
- Very willing

19. What features would you like to see in such an app? (Optional)

- _____

20. Why would or wouldn't you use the website/app described in the previous question? (Optional)

- _____

Survey Results

Our survey results can be found [here](#).

Appendix C: Usability Testing Script

User Tasks

1. Users should be able to view their all their projects and upcoming meetings so that they can keep track of their schedule and project progress.
2. Users should be able to create and schedule a new meeting to discuss project matters with their groupmates.
3. Users should be able to join and participate in a meeting, where they can get support from the AI assistants to keep their meeting productive and organized.
4. Users should be able to access files and documents related to each project, including past meeting summaries and transcripts for reference.

Usability Testing Tasks

Task 1

1. You just launched the app and you would like to see what your schedule is like for the upcoming week.

Task 2

1. You would like to schedule a meeting for your business FYP project to discuss presentation plans with your groupmates in the near future.
2. You have not yet settled on a date with your groupmates and would like to quickly find an optimal time where everyone is free.
 - a. Note: This is not your first time scheduling a meeting on Promptly, and all your groupmates have already set up their accounts and linked their calendars.

Task 3

1. Your group meeting for your business FYP presentation is happening now and you want to join the meeting.
2. Before joining the meeting you want to make sure that the AI meeting assistants are set up to support the group discussion.
3. During the meeting, your mind wandered and you lost track of what was just mentioned. You would like some help from the AI assistants to help you get back on track.
4. As the meeting went on, you notice that the discussion has stalled and there is awkward silence. You would like to get some help from the AI assistants on how to move the discussion forward. In particular, you would like some suggestions on how to structure your presentation.
5. After much help from the AI assistants, you want to include it in the summary(edit the summary notes)
6. You notice that some groupmates are quieter than usual. To keep the discussion rich, you want to probe some of the quieter groupmates for some of their opinions.
7. After an hour of discussion, you have settled on the plan moving forward for your presentation. You would like to exit the meeting and save what was discussed during it.

Task 4

1. A few days have passed since the meeting. As you are working on the business FYP presentation, you want to reference some of the points discussed during the previous meeting with your groupmates on the project focus.

Post-interview questions

1. What features/components do you like for each task/screen?
2. What features/components do you dislike about each task/screen, and why? How can we improve these features/components?
3. Rate the following statements on your experience with Promptly on a scale of 1 to 5 (1 – strongly disagree, 5 – strongly agree)
 - a. I found the system's user interface to be intuitive and easy to use.
 - b. I found the process of completing the tasks I want in the system to be straightforward.
 - c. I found the system's features to be helpful in scheduling and hosting group meetings.

References

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2. Ruhleder, K., & Jordan, B. (2001). Co-constructing non-mutual realities: Delay-generated trouble in distributed interaction. *Computer Supported Cooperative Work (CSCW)*, 10(1), 113–138. <https://doi.org/10.1023/A:1011243905593>
3. Arslan, M., Munawar, S., & Cruz, C. (2024). Business insights using RAG–LLMs: a review and case study. *Journal of Decision System*, 1–30. <https://doi.org/10.1080/12460125.2024.2410040>